

EPI TRIMBAKESHWAR - MEP SERVICES, STP AND ENVIRONMENTAL ENGINEERING

Design Criteria: Sustainable Campus Model with 100% Water Recycling and Green Standards

Power Supply: 11kV Dedicated Feeder from MSEDCL with 315/500 kVA Step-down Substation

STP Technology: Moving Bed Biofilm Reactor (MBBR) / SBR 60 KLD capacity with tertiary filtration

Water Requirement: 135 LPCD for resident students and staff; 45 LPCD for day scholars/admin

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1. ELECTRICAL DISTRIBUTION: Underground XLPE armored cabling, MCB/ELCB distribution boards, energy-efficient LED.
 2. BACKUP POWER: 125/160 kVA acoustic enclosed Diesel Generator set for critical loads, hostel pumps, and lighting.
 3. WATER SUPPLY: Borewells + municipal supply; Underground Domestic Sump (1.5 Lakh L) + Overhead Tanks (50,000 L).
 4. STP RECYCLING: Treated effluent meeting CPCB standards utilized for toilet flushing and campus horticulture.
 5. RAINWATER HARVESTING: Rooftop catchment channeled through silt traps into 6 percolation/recharge borewells.
 6. FIRE PROTECTION: Wet riser system, yard hydrants, hose reels, manual call points, and NBC 2016 Part 4 compliant.
 7. SOLAR PHOTOVOLTAIC: Rooftop grid-interactive solar plant and 5,000 LPD solar thermal water heating for hostels.